The Map in Your Head

How brains — yours, a rat's, a bat's — know exactly where they are

Pre-production pack · matsapphouse short science film · Prepared Nov 4, 2026 by Grok Bot
Topic: neuroscience (spatial navigation), with an animal-kingdom thread (rats, flying bats)
Target length: about 4 minutes (≈560 narration words at ~140 wpm) · Status: **ready for Mat to check**. No images, video or music generated.

1. Logline & approach

Every time you walk to the station without checking your phone, a hidden map is switching on inside your skull. This film follows the discoveries — from a rat in a London lab in 1971 to bats flying through a room in Israel and London cabbies in an MRI scanner — that revealed the brain's "inner GPS", which won the 2014 Nobel Prize.

Look: pink and green palette. Pink glows = cells firing; green = the world/map. Dark backgrounds with glowing dots, hexagon grids, gentle motion graphics. **Tone:** warm, curious, a little playful. **Narrator:** one calm voice.

2. Narration script (with sources)

Bracketed numbers point to the source list in section 5. Times are approximate.

0:00 – Cold open

> Close your eyes. Now point to your front door.
> You probably can. You didn't check a map. You didn't ask your phone. Somewhere inside your head, a map was already open.
> So where is it? And who drew it?

0:20 – The question

> For centuries, philosophers argued about how we sense space. In the 1940s, the psychologist Edward Tolman watched rats learn mazes and guessed they were building a "cognitive map" inside their brains [1]. But nobody could see it.

0:45 – Place cells (1971)

Then, in 1971, John O'Keefe in London listened in on single nerve cells in a rat's hippocampus — a curled, seahorse-shaped structure deep in the brain [1][2].
He noticed something strange. One cell fired only when the rat was in one particular spot in the room. Move the rat somewhere else, and a different cell lit up.
He called them place cells. Together, they formed a map of the room — one spot, one cell [1].

1:25 – Head-direction cells (1990)

But a map is useless without a compass. In 1990, Jeffrey Taube, Robert Muller and James Ranck described cells that fire depending on which way the rat's head is pointing — no matter where it is [3]. Turn a landmark on the wall, and the cells' "north" turns with it [4]. The brain carries its own compass.

1:55 – Grid cells (2005)

More than three decades after place cells, May-Britt and Edvard Moser in Norway were recording next door to the hippocampus, in the entorhinal cortex. There they found cells that fired at many places — and when they plotted those places, they made a pattern of hexagons, like a honeycomb laid across the floor [1][5].
Grid cells. The brain's own graph paper. A coordinate system for measuring where you are and how far you've gone.
In 2014, O'Keefe and the Mosers shared the Nobel Prize for discovering the brain's positioning system [1].

2:40 – Bats in three dimensions

But rats live on the floor. What about animals that fly?
In 2013, Michael Yartsev and Nachum Ulanovsky fitted Egyptian fruit bats with tiny wireless recorders and let them fly around a room [6]. Their place cells didn't mark spots on a floor — they marked bubbles of space in the air. Up, down, left, right: all three directions were mapped with similar precision [6].

3:10 – And in us

Humans have this map too. Patients having brain surgery for epilepsy played a navigation video game while doctors recorded their neurons — and found grid-like patterns, much like the rat's honeycomb [7].
And the map can grow. London taxi drivers, who memorise the city's tangled streets, had a larger rear part of the hippocampus than people who didn't drive taxis — and the longer they'd driven, the bigger it was [8].

3:40 – Why it matters / close

The hippocampus and entorhinal cortex are often among the first areas hit by Alzheimer's disease — one reason people with it can lose their way [1]. Understanding the map may help us protect it.
So next time you find your way home in the dark, thank a few thousand cells: a compass, a sheet of graph paper, and a map, quietly drawn — one place at a time.
Title card: The Map in Your Head.

Fact-check notes: "seahorse-shaped" is the conventional description of the hippocampus' name. Taxi study: 16 drivers vs 50 controls; overall hippocampal volume was NOT larger — only the posterior part (the anterior part was smaller) [8]. Script wording says "rear part" to stay accurate. Human grid study: 14 patients [7].

3. Shot-by-shot storyboard (text)

#	Time	Visual	Motion / sound	On-screen text
1	0:00	Black screen; a soft pink dot pulses in the centre.	Silence, then a single piano note per pulse.	—
2	0:06	Silhouette of a person, eyes closed, arm pointing off-frame.	Slow push-in.	—
3	0:12	Camera pulls back into the head: glowing green streets appear inside the skull outline.	Whoosh; song intro starts low.	—
4	0:20	Old-paper style maze, a cartoon rat running it; thought bubble with a tiny map.	Sepia, light film grain.	"Cognitive map" – Tolman, 1940s
5	0:45	Top-down view of a square room; rat walks; brain inset with the hippocampus highlighted green.	Hippocampus gently glows.	Hippocampus
6	0:55	Rat reaches a corner: a pink dot "pops" on the floor there. Click sound = spike.	Clicks synced to pops.	—
7	1:05	More dots of different pink shades build up until the floor is covered.	Timelapse; clicks layer.	"Place cells" – O'Keefe, 1971
8	1:25	Rat in a round arena, a white card on the wall; a compass needle floats above its head.	Rat turns, needle tracks head direction.	"Head-direction cells" – 1990
9	1:42	The card rotates around the wall; the compass "north" rotates with it.	Smooth 90° rotation.	—
10	1:55	Map of Norway → zoom to Trondheim lab; brain inset, entorhinal cortex in pink.	Quick map zoom.	Entorhinal cortex
11	2:05	Rat wanders; pink dots appear at many spots — then green lines join them into hexagons.	Satisfying "snap" as hexagons form.	"Grid cells" – Moser & Moser, 2005
12	2:25	Three portrait silhouettes in pink/green; gold medal shape.	Gentle chord.	Nobel Prize 2014
13	2:40	Fruit bat takes off in a dark room; tiny backpack recorder blinks green.	Wing flaps; song verse 2 enters.	Egyptian fruit bat
14	2:52	3D wireframe room; glowing pink bubbles hang in mid-air where cells fire as the bat flies through.	Camera orbits the room.	"Place fields in 3D" – 2013
15	3:10	Simple video-game maze on a hospital-room monitor (no faces, no gore).	Beep of monitor, soft.	Humans, 2013
16	3:20	Hexagon pattern from shot 11 fades over the game map.	Match cut.	—
17	3:25	Black London cab driving through a tangle of green streets; hippocampus inset, rear part swelling slightly pink.	Traffic ambience.	London taxi drivers

18	3:40	The glowing map inside the head flickers and dims at the edges.	Music thins.	—
19	3:50	Person walking home at night, a faint pink compass, green grid and dots trailing behind them.	Song chorus swells.	—
20	4:05	Title card on pink-to-green gradient.	Last chorus line, fade.	The Map in Your Head · sources in description

4. Original song: "Honeycomb Floor"

Style: dreamy indie-pop, 92 bpm, soft synth arpeggio (like ticking spikes), warm bass, light female or male vocal. Fits under narration: verses quiet, chorus over shots 11, 19–20.

[Intro]
Click… click…
Somewhere in the dark
a little light says "here"

[Verse 1]
Shut my eyes on the platform line
still know the way, still know it's mine
a quiet lamp in a curling sea
one small spark for every place I'll be

[Pre-Chorus]
Turn my head and a needle swings
north is a feeling, not a thing

[Chorus]
I've got a honeycomb floor
laid out under my feet
pink lights, green streets
every corner I meet
no screen, no star, no door
I still find my way home
there's a map in my head
and I'm never alone

[Verse 2]
Fruit bat spinning through the midnight air
lights up bubbles hanging everywhere
up and over, left and right
a three-way map drawn in flight

[Pre-Chorus]
Cabbies learning every lane
the map grows bigger in the brain

[Chorus]
I've got a honeycomb floor
laid out under my feet
pink lights, green streets
every corner I meet
no screen, no star, no door
I still find my way home
there's a map in my head
and I'm never alone

[Bridge]
And if the lights go dim
and the edges fade
I'll hold the hand
of the one who stayed
one place at a time
one cell at a time

[Final Chorus / Outro]
Honeycomb floor…
pink lights, green streets
there's a map in my head
one place at a time

Lyrics are original and written for this film. Production note: no music has been generated (SuperGrok Imagine usage is nearly out) — make it later with whichever free tool Mat prefers.

5. Sources (opened and checked Nov 4, 2026)

1. Nobel Assembly at Karolinska Institutet, 2014 Nobel Prize in Physiology or Medicine press release (Tolman's cognitive map; O'Keefe 1971 place cells; Mosers 2005 grid cells, hexagonal grid; human evidence; Alzheimer's link). https://www.nobelprize.org/prizes/medicine/2014/press-release/
2. O'Keefe J. & Dostrovsky J. (1971) The hippocampus as a spatial map. Brain Research 34:171–175 (cited in [1]).
3. Taube J.S., Muller R.U., Ranck J.B. Jr (1990) Head-direction cells recorded from the postsubiculum in freely moving rats. I. J Neurosci 10:420–435. https://www.jneurosci.org/content/10/2/420
4. Taube, Muller & Ranck (1990) … II. Effects of environmental manipulations (cue-card rotation rotates preferred direction). J Neurosci 10:436–447. https://www.jneurosci.org/content/10/2/436
5. Hafting T. et al. (2005) Microstructure of a spatial map in the entorhinal cortex. Nature 436:801–806 (cited in [1]).
6. Yartsev M.M. & Ulanovsky N. (2013) Representation of three-dimensional space in the hippocampus of flying bats. Science 340:367–372. https://pubmed.ncbi.nlm.nih.gov/23599496/
7. Jacobs J. et al. (2013) Direct recordings of grid-like neuronal activity in human spatial navigation. Nature Neuroscience 16:1188–1190. https://pmc.ncbi.nlm.nih.gov/articles/PMC3767317/
8. Maguire E.A. et al. (2000) Navigation-related structural change in the hippocampi of taxi drivers. PNAS 97:4398–4403. https://pmc.ncbi.nlm.nih.gov/articles/PMC18253/

Before production — for Mat: pick narrator voice; OK the title and song; decide whether to name "Jeffrey Taube, Robert Muller and James Ranck" or simplify to "a team in New York". Nothing has been published anywhere.